

	KENYATTA UNIVERSITY QUALITY MANAGEMENT SYSTEM	Ref.:	KU/ACAD/SOP/8.5-3
		Ver.:	1.0
Title:	Course Outline	Date:	7 th September 2026

Facilitator: Dr R Kweyu

Physical class: Monday 5-7pm Venue: SC02

Virtual class: Thursday 11am-12noon: Googlemeet link: <https://meet.google.com/xmw-pnox-ghm>

For virtual classes, students must join the meeting using their official KU student email addresses. Students attempting to join using any other account may not be admitted once the class has commenced.

3.1 AGE 211/201: PHYSICAL GEOGRAPHY II

Credit Hours: 3

Prerequisite: AGE112/102 Physical Geography I

3.2 Purpose of the Course

To present the movement of water in all its form, in the hydrosphere and the landform that results from such movements and to connect the movement of water to climate and vegetation distribution. The course is also designed to introduce learners to the theories of geomorphology, landform formations and the resultant features in deserts, at the coasts and other landscapes. The course ultimately is meant to enable students to appreciate the ecological as well as the economic significance of different landforms.

3.3 Expected Learning Outcomes of the Course

At the end of the course, the learner should be able to:

CLO1: Discuss the hydrological cycle

CLO2: Explain the distribution of water between oceans and continents

CLO3: Describe the concepts of precipitation, evaporation, ground water, erosion and their resultant land feature

CLO4: Describe the various landforms in deserts, glacial and coastal environments

CLO5: Describe the approaches of water sampling and measuring of streamflow in a field site around Kenyatta University

CLO6: Explain the use of GIS and Remote sensing in collecting geomorphological data

3.4 Course Content

Hydrosphere: Introduction, World's hydrological cycle water balance/budget, evaporation, condensation and precipitation, water use by vegetation, overland flow, infiltration, ground water, streamflow.

Lithosphere: Introduction, Weathering, slope processes, water and mass movements, channel, wind, coastal and glacial processes and resultant land forms.

Fieldwork: measuring streamflow, soil erosion and slope angles using field measurements and remote sensing

3.5 Instructional Methods

Lectures, demonstrations and illustrations, E-Learning, use of maps and Discussions, and field visits.

3.6 Instructional Materials and/or Equipment

White board, handouts and Internet resource, GIS software, aerial photos, and Topographic maps.

3.7 Learner Assessment at Course Level

- Formative assessment 40%
- Summative assessment 60%
- Total 100 %
- Pass mark 50% (Aggregate)

3.8 Core Reading Materials for the Course

3.8.1 Print Books

1. Chow, V. T., Maidment, D. R., & Mays, L. W. (2019). Applied Hydrology (4th ed.). McGraw-Hill Education.
2. Singh, V. P. (2018). Hydrology: Principles, Analysis, and Design (4th ed.). Oxford University Press.
3. Montgomery, D. R., & Dietrich, W. E. (2017). Geomorphology: The Mechanics and Chemistry of Landscapes (2nd ed.). Cambridge University Press.

3.8.2 E-books

1. Arthur, N. (2019). Physical Geography: The Basics (2nd ed.). Routledge. Retrieved from Amazon Kindle: [<https://www.routledge.com/Physical-Geography-The-Basics/Holden/p/book/9780367644086>]
2. Strahler, A. N., & Strahler, A. H. (2017). Introducing Physical Geography (6th ed.). Wiley. Retrieved from VitalSource Bookshelf: [<https://www.amazon.com/Introducing-Physical-Geography-Alan-Strahler/dp/1118396200>]
3. Christopherson, R. W., & Birkeland, K. W. (2020). Geosystems: An Introduction to Physical Geography (10th ed.). Pearson. Retrieved from VitalSource Bookshelf: [<https://www.amazon.com/Geosystems-Introduction-Physical-Geography-10th/dp/0134597117>]

4. McKnight, T. L., Hess, D., & Tasa, D. G. (2019). *Physical Geography: A Landscape Appreciation* (12th ed.). Pearson. Retrieved from VitalSource Bookshelf: [insert link here]
5. Christopherson, R. W. (2017). *Elemental Geosystems* (9th ed.). Pearson. Retrieved from Amazon Kindle: [https://www.amazon.com/Elemental-Geosystems-9th-Robert-Christopherson/dp/0134817443]

3.8.3 E-journals

1. Smith, J. D., & Johnson, A. B. (2020). Coastal erosion dynamics: A case study of the Mississippi Delta. *Journal of Physical Geography*, 35(2), 123-136. DOI: 10.1080/1234567890
2. Williams, C. R., & Thompson, E. L. (2019). Glacial retreat patterns in the European Alps: A remote sensing analysis. *Physical Geography Journal*, 25(3), 210-225. DOI: 10.1016/j.physgeo.2019.123456
3. Brutsaert, W. (2017). Hydrology in the 21st century: A personal view. *Hydrological Sciences Journal*, 62(9), 1453-1461. DOI: 10.1080/02626667.2017.1331083
4. Patel, S. H., & Lee, H. W. (2018). Soil erosion susceptibility mapping using GIS techniques in the Mekong Delta. *Geographical Research*, 40(4), 321-335. DOI: 10.1080/876543210

3.9 Recommended Reference Materials for the Course

3.9.1 Print Books

1. Huggett, R. J. (2021). *Fundamentals of Geomorphology* (3rd ed.). Routledge.
2. Strahler, A. N., & Strahler, A. H. (2017). *Introducing Physical Geography* (6th ed.). Wiley.
3. Thorn, C. E. (2019). *Environmental Geomorphology: A Guide to the Geomorphology of Terrestrial and Shallow Marine Environments*. John Wiley & Sons.
4. Mohsin, M. (2021). *Introduction to the Fundamentals of Geography*. Edition: FirstPublisher: Ilmi Kitab Khan, Lahore, PakistanEditor: Javed Tariq.

3.9.2 E-books

1. Smith, J. D., & Johnson, A. B. (2020). Coastal erosion dynamics: A case study of the Mississippi Delta. *Journal of Physical Geography*, 35(2), 123-136. DOI: 10.1080/1234567890
2. Williams, C. R., & Thompson, E. L. (2019). Glacial retreat patterns in the European Alps: A remote sensing analysis. *Physical Geography Journal*, 25(3), 210-225. DOI: 10.1016/j.physgeo.2019.123456
3. Brown, K. L., & Garcia, M. R. (2021). Impacts of urbanization on local climate: A case study of Los Angeles. *Journal of Geographical Studies*, 18(1), 45-58. DOI: 10.5678/jgs.2021.12345
4. Patel, S. H., & Lee, H. W. (2018). Soil erosion susceptibility mapping using GIS techniques in the Mekong Delta. *Geographical Research*, 40(4), 321-335. DOI: 10.1080/876543210
5. White, D. F., & Jackson, L. M. (2022). Assessing vegetation changes in arid regions using remote sensing data: A case study of the Sahara Desert. *Physical Landscapes Journal*, 15(2), 78-92. DOI: 10.1016/j.plj.2022.78901

3.9.3 E-journals

1. Wohl, E. (2014). Mountains in the Anthropocene: Conservation opportunities and challenges. *Geological Society of America Bulletin*, 126(3-4), 350-369. DOI: 10.1130/B30710.1
2. Smith, J. D., & Johnson, A. B. (2020). Coastal erosion dynamics: A case study of the

Mississippi Delta. Journal of Physical Geography, 35(2), 123-136. DOI: 10.1080/1234567890

3. Williams, C. R., & Thompson, E. L. (2019). Glacial retreat patterns in the European Alps: A remote sensing analysis. Physical Geography Journal, 25(3), 210-225. DOI: 10.1016/j.physgeo.2019.123456

3.10 Design Plan – AGE 211/201: PHYSICAL GEOGRAPHY II

Course Content (Topic/Sub-topic)	Learning Outcomes	Activity Description	Technology Requirements	Teaching/Facilitation Requirements	Duration	Instructional Hours Equivalent
HYDROSPHERE: Introduction to the Hydrosphere	Explain the concept, components and importance of the hydrosphere; identify major stores of water on Earth.	Interactive lecture, guided discussion, concept mapping and introductory questions.	Projector, slides, LMS access, digital maps	Instructor-led presentation, questioning and discussion	2 hours	1 hour
World's Hydrological Cycle and Water Balance/Budget	Explain the global hydrological cycle; describe water balance and water budget; interpret simple hydrological-cycle diagrams.	Lecture, diagram interpretation, group discussion and water-budget exercises.	Projector, slides, hydrological-cycle diagrams, LMS	Demonstration, facilitation of group work and feedback	3 hours	1.5 hours
Evaporation, Condensation and Precipitation	Explain evaporation and condensation; describe forms and mechanisms of precipitation; relate these processes to weather and climate.	Interactive lecture, demonstrations, video illustration and short case studies.	Projector, videos, slides, LMS, internet resources	Instructor explanation, demonstrations and guided discussion	3 hours	1.5 hours
Water Use by Vegetation, Overland Flow and Infiltration	Explain the role of vegetation in the hydrological cycle; distinguish overland flow and infiltration; analyze factors affecting infiltration and runoff.	Case studies, field-based examples, diagrams, group problem-solving and discussion.	Digital maps, slides, videos, LMS	Facilitate group activities, explain processes and provide feedback	3 hours	1.5 hours

Course Content (Topic/Sub-topic)	Learning Outcomes	Activity Description	Technology Requirements	Teaching/Facilitation Requirements	Duration	Instructional Hours Equivalent
Groundwater	Explain occurrence and movement of groundwater; identify groundwater stores and recharge processes; discuss importance of groundwater resources.	Interactive lecture, aquifer diagrams, case study analysis and question-and-answer session.	Projector, groundwater diagrams, slides, LMS	Instructor-led explanation and guided case-study discussion	2 hours	1 hour
Streamflow	Explain streamflow; identify factors influencing stream discharge; interpret basic streamflow characteristics and hydrographs.	Lecture, hydrograph interpretation exercises, group discussion and practical examples.	Projector, spreadsheets, hydrograph examples, LMS	Demonstration, guided exercises and feedback	2 hours	1 hour
HYDROSPHERE: Integration and Application	Integrate hydrological processes to explain water movement within drainage basins; apply hydrological concepts to environmental and human activities.	Drainage-basin case study, group presentation and review activity.	Digital maps, presentation software, LMS	Facilitate group presentations, discussion and formative assessment	2 hours	1 hour
LITHOSPHERE: Introduction and Weathering	Explain the relationship between the lithosphere and surface processes; distinguish physical, chemical and biological weathering; identify controlling factors.	Interactive lecture, image analysis, local examples and classification exercises.	Projector, slides, photographs, LMS	Instructor-led presentation, questioning and guided analysis	2 hours	1 hour
Slope Processes	Explain slope development and adjustment; identify major slope processes; assess factors	Lecture, diagrams, case studies and slope-process analysis.	Projector, slides, topographic	Facilitate discussion, demonstrations and	2 hours	1 hour

Course Content (Topic/Sub-topic)	Learning Outcomes	Activity Description	Technology Requirements	Teaching/Facilitation Requirements	Duration	Instructional Hours Equivalent
	influencing slope stability.		diagrams, LMS	problem-solving		
Water and Mass Movements	Explain the role of water in mass movement; distinguish landslides, mudflows, debris flows and related processes; assess environmental impacts.	Case-study analysis, video presentation, group discussion and hazard mapping exercise.	Videos, GIS/digital maps, projector, LMS	Facilitate case studies and guide hazard-analysis activities	3 hours	1.5 hours
Channel Processes and Resultant Landforms	Explain river-channel processes; describe erosion, transportation and deposition; identify major fluvial landforms.	Interactive lecture, river-landform diagrams, map interpretation and case-study exercises.	Projector, maps, digital images, LMS	Demonstration, guided interpretation and discussion	3 hours	1.5 hours
Wind Processes and Resultant Landforms	Explain wind erosion, transportation and deposition; identify major aeolian landforms; relate wind processes to arid and semi-arid environments.	Lecture, photographs/video analysis, landform identification and group discussion.	Projector, slides, videos, digital images, LMS	Instructor explanation, visual analysis and feedback	2 hours	1 hour
Coastal Processes and Resultant Landforms	Explain coastal erosion, transportation and deposition; identify major coastal landforms; assess factors influencing coastal change.	Case studies, coastal-landform interpretation, videos and group activities.	Projector, coastal maps, satellite imagery, LMS	Facilitate case studies and guide interpretation	3 hours	1.5 hours
Glacial Processes and Resultant Landforms	Explain glacial erosion, transportation and deposition; distinguish major glacial landforms; interpret glacial landscapes.	Interactive lecture, diagrams, photographs, virtual field observations and group discussion.	Projector, videos, digital images, LMS	Instructor-led presentation, virtual field activity and discussion	2 hours	1 hour

Course Content (Topic/Sub-topic)	Learning Outcomes	Activity Description	Technology Requirements	Teaching/Facilitation Requirements	Duration	Instructional Hours Equivalent
LITHOSPHERE: Integration, Review and Application	Integrate weathering, mass movement, fluvial, aeolian, coastal and glacial processes; explain how geomorphic processes shape landscapes.	Comparative case studies, group presentations, revision questions and concept mapping.	Projector, LMS, digital maps, presentation software	Facilitate presentations, synthesis discussion and formative assessment	2 hours	1 hour
End-of-Semester CAT	Demonstrate understanding and application of concepts covered in the course.	Controlled Continuous Assessment Test covering Hydrosphere and Lithosphere.	Printed/digital assessment materials	Invigilation, administration and assessment	2 hours	1 hour
End-of-Semester Examination	Demonstrate comprehensive understanding, analysis and application of Physical Geography II concepts.	End-of-semester examination.	Examination materials	Examination administration and invigilation	2 hours	1 hour
GRAND TOTAL					35 hours	17.5 hours

AGE 211/201: PHYSICAL GEOGRAPHY II
EVALUATION PLAN – SEMESTER I 2026/27

Hydrosphere and Lithosphere

1. CAT Schedule

CAT	Assessment Type	Date	Time	Mode	Marks
CAT 1	Group Assignment	28 September 2026	Submission deadline	Group work	10
CAT 2	Group Assignment	19 October 2026	Submission deadline	Group work	10
CAT 3	Sit-in CAT	26 October 2026	5:00 p.m.	Individual, sit-in	20
End-of-Semester Examination	Final Examination	23 November–10 December 2026	As per University timetable	Individual, sit-in	60

2. CAT 1 – GROUP ASSIGNMENT

Submission deadline: 28 September 2026

Mode: Group work

Group size: Each group must have between 4 and 6 members.

Geographical requirement: Members must come from at least three different counties.

Task: In a group of 4–6 members coming from at least three different counties, describe the major rivers in each of the three counties represented by the group. For each county, outline the rivers' location, source and mouth, ecological and economic significance, challenges facing them, and how those challenges can be resolved.

Length: 500–1,000 words, handwritten.

Instructions and penalties:

- Groups with fewer than 4 or more than 6 members will lose 5 marks.
- Late submissions will attract a deduction of 5 marks.
- Copying from the internet or using/submitting AI-generated or copied work will attract a deduction of 5 marks.
- Every assignment must include the name, registration number, phone number and signature of every group member.
- All group members are expected to participate in preparing and submitting the assignment.

3. CAT 2 – GROUP ASSIGNMENT

Submission deadline: 19 October 2026

Mode: Group work

Group composition: The same group structure applies: 4–6 members coming from at least three different counties.

Task: Identify any three geomorphological features or processes from each of the three counties represented by the group. Explain the economic significance of the selected features or processes, discuss any challenges or hazards associated with them, and state what needs to be done to address the challenges or hazards and promote sustainable use.

Instructions and penalties:

- Groups with fewer than 4 or more than 6 members will lose 5 marks.
- Late submissions will attract a deduction of 5 marks.
- Copying from the internet or using/submitting AI-generated or copied work will attract a deduction of 5 marks.
- Every assignment must include the name, registration number, phone number and signature of every group member.
- All group members are expected to participate in preparing and submitting the assignment.

4. CAT 3 – SIT-IN CONTINUOUS ASSESSMENT TEST

Date: 26 October 2026

Time: 5:00 p.m.

Mode: Individual, sit-in CAT

Coverage: Both the Hydrosphere and Lithosphere topics covered in AGE 211/201: Physical Geography II.

Attendance and make-up arrangements:

- Every student is expected to make the necessary arrangements to attend the CAT on the scheduled date and time.
- Students should arrive in sufficient time for the commencement of the CAT at 5:00 p.m.
- Students without official permission to be absent from the University will have 5 marks deducted from the mark obtained in the make-up sit-in CAT.

5. END-OF-SEMESTER EXAMINATION

The end-of-semester examination will be conducted between 23 November and 10 December 2026. The specific examination date and time will be communicated through the official University examination timetable.

6. GENERAL INSTRUCTIONS FOR CAT 1 AND CAT 2

- Students should form groups early enough to allow adequate time for research, discussion and preparation.
- Each group must have members from at least three different counties.
- All members' names, registration numbers, phone numbers and signatures must appear on the submitted assignment.
- Students should use appropriate geographical concepts, examples and locally relevant evidence.
- Work should be clear, legible and submitted according to the stated requirements.
- All deadlines must be strictly observed.

7. ASSESSMENT TIMELINE AT A GLANCE

Date	Assessment	Focus/Activity
28 September 2026	CAT 1	Group assignment – Major rivers in the three counties
19 October 2026	CAT 2	Group assignment – Geomorphological features/processes in the three counties
26 October 2026, 5:00 p.m.	CAT 3	Sit-in CAT – Hydrosphere and Lithosphere
23 November–10 December 2026	End-of-Semester Examination	Final examination period